



Bo-Yuan Ning 宁博元

Curriculum Vitae

Working Experience

- 2023–present **Research Faculty**, *SINAP*, Chinese Academy of Science.
- 2021–2022 **Postdoc Fellowship**, *Institute of Modern Physics*, Fudan University.
Advisor: Prof. Xi-Jing Ning (宁西京)
- 2016–2020 **Postdoc Fellowship**, *HPSTAR*, Chinese Academy of Engineering Physics.
Advisor: Prof. Tsu-Chien Weng (翁祖谦)
- 2015–2016 **Research Assistant**, *Institute of Modern Physics*, Fudan University.
Advisor: Prof. Xi-Jing Ning (宁西京)

Education

- 2009–2015 **Ph.D. in Physics**, *Fudan University*.
Ph.D dissertation: **Quantum Control on Spin-1 Bose-Einstein Condensation**,
Supervisors: Prof. Wenxian Zhang (张文献) & Prof. Jun Zhong (庄军),
Major: Atomic, Molecular and Optical Physics,
Department of Optical Science and Engineering
- 2005–2009 **B.S. in Physics**, *University of Shanghai for Science and Technology*.
B.S. thesis: **How to realize a macroscopic quantum single pendulum?**,
Supervisor: Prof. Junshan Ma (马军山),
Major: Optics Information of Science and Technology,
School of Optical Electrical and Computer Engineering

Research Fields and Interests

I. Statistical Mechanics Methods for Thermodynamic Computations of Condensed-Matter Materials

- * Development of two ensemble-theory based methods, direct integral approach (DIA) and effective entropy approach (EEA), to solve the many-body partition function and compute the thermodynamic properties of solid-state materials and amorphous liquids in a combination with quantum first-principle calculations.

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II. Phase Transition and Stability of Condensed-Matter Materials under High Pressure-Temperature Conditions

- * Applications of DIA to investigate the structural phase transitions of metallic systems, Vanadium, Aluminium, Zirconium, Gold up to 600 GPa.
- * Applications of DIA and EEA to investigate the structural stability of large molecules up to 2500K.
- * Applications of DIA to investigate the optimal temperature condition for the growth of 2-dimensional materials, γ -gryphyne and silicene.

III. Equations of State of Condensed-Matter Materials under Extreme Conditions

- * Applications of DIA to investigate the isothermal and isobaric equations of state of (solid) Vanadium, Aluminium, Zirconium, Gold, and (solid and liquid) Copper and Argon up to 600 GPa and 3000K.

IV. Nonequilibrium Statistical Methods for Amorphous-Crystal Materials

- * Using molecular dynamics (MD) simulations and applications of nonequilibrium statistical model (single-atom statistical method) to predict the glass forming ability and critical cooling rate of metallic glass. (In collaborations with Southern University of Science and Technology)

V. Previous Research Experiences

- * *Spin Dynamics in Ultracold Bose-Einstein Condensation*. Development of two control methods to independently manipulate the spin-exchange and magnetic dipolar-dipolar interactions in an ultracold quantum spinor condensate.
- * *Ultrafast Spectrum*. Using femtosecond laser and pump-probe technique to analyze the spectrum of ultrafast processes. (In charge of the maintenance of the ultrafast spectrum lab in HPSTAR simultaneously)
- * *Pulse Laser Deposition*. Investigations of electric current effects on growing high-quality metallic thin film (Cu, Ni) by nanosecond pulse laser deposition method.

Publications

- * Equation of state for tungsten predicted by ensemble theory, Yue-Yue Tian, **Bo-Yuan Ning**, X.-D. Xiang, Hui-Fen Zhang*, Xi-Jing Ning[†], **arXiv:2210.16736** (2022) [*preprint*].
- * An *ab initio* study of structural phase transitions of crystalline aluminium under ultrahigh pressures based on ensemble theory, **Bo-Yuan Ning***, Li-Yuan Zhang, Vol.**218**, No.**111960**, **Computational Materials Science** (2023).
- * Pressure-induced structural phase transitions of zirconium: An *ab initio* study based on statistical ensemble theory, **Bo-Yuan Ning***, Vol.**34**, No.**505402**, **Journal of Physics: Condensed Matter** (2022).
- * Pressure-induced structural phase transition of vanadium: A revisit from the perspective of ensemble theory, **Bo-Yuan Ning**, Xi-Jing Ning*, Vol.**34**, No.**425404**, **Journal of Physics: Condensed Matter** (2022).

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- * Efficient approaches to solutions of partition function for condensed matters, **Bo-Yuan Ning**, Le-Cheng Gong, Tsu-Chien Weng*, Xi-Jing Ning[†], Vol.**33**, No.**115901**, **Journal of Physics: Condensed Matter** (2021).
- * Which phase of Ta₂O₅ being of the largest dielectric constant?, Hui-Fen Zhang, **Bo-Yuan Ning**, Tsu-Chien Weng, Xi-Jing Ning*, Vol.**104**, No.**6413**, **Journal of American Ceramics Society** (2021).
- * How accurate for phonon models to predict the thermal dynamical properties of crystals?, Le-Cheng Gong, **Bo-Yuan Ning**, Chen Ming, Tsu-Chien Weng, Xi-Jing Ning*, Vol.**33**, No.**085901**, **Journal of Physics: Condensed Matter** (2020).
- * Comparison of two efficient methods for calculating partition functions, Le-Cheng Gong, **Bo-Yuan Ning**, Tsu-Chien Weng, Xi-Jing Ning*, Vol.**21**, No.**1050**, **Entropy** (2019).
- * A new model to predict optimum conditions for growth of 2D materials on a substrate, Yu-Peng Liu, **Bo-Yuan Ning**, Le-Cheng Gong, Tsu-Chien Weng, Xi-Jing Ning*, Vol.**9**, No.**978**, **Nanomaterials** (2019).
- * What retards the response of graphene based gaseous sensor, Hui-Fen Zhang, **Bo-Yuan Ning**, Tsu-Chien Weng, Dong-Ping Wu, Xi-Jing Ning*, Vol.**295**, No.**188**, **Sensors and Actuators A: Physical** (2019).
- * A diffusion model for solute atoms diffusing and aggregating in nuclear structural materials, Quan Song, Fan-Xin Meng, **Bo-Yuan Ning**, Jun Zhuang, Xi-Jing Ning*, Vol.**26**, No.**126601**, **Chinese Physics B** (2017).
- * Rapidly calculating the partition function of macroscopic systems, Jing-Tian Li, **Bo-Yuan Ning**, Le-Cheng Gong, Jun Zhuang, Xi-Jing Ning*, Vol.**26**, No.**030501**, **Chinese Physics B** (2017).
- * Electronic band structure and sub-band-gap absorption of Nitrogen hyperdoped Silicon, Zhen Zhu, Hezhu Shao, Xiao Dong, Ning Li, **Bo-Yuan Ning**, Li Zhao, Jun Zhuang*, Vol.**5**, No.**10513**, **Scientific Reports** (2015).
- * Site-selective substitutional doping with atomic precision on stepped Al (111) surface by single-atom manipulation, Chang Chen, Jinhu Zhang, Guofeng Dong, Hezhu Shao, **Bo-Yuan Ning**, Li Zhao, Xi-Jing Ning, Jun Zhuang*, Vol.**9**, No.**235**, **Nanoscale Research Letters** (2014).
- * Hybrid functional studies on impurity-concentration-controlled band engineering of Chalcogen-Hyperdoped silicon, Hezhu Shao, Cong Liang, Zhen Zhu, **Bo-Yuan Ning**, Xiao Dong, Xi-Jing Ning, Li Zhao, Jun Zhuang*, Vol.**6**, No.**085801**, **Applied Physics Express** (2013).
- * Reliable lateral atom manipulation by thermal activation on metal fcc(001) surfaces, Jinhu Zhang, Chang Chen, **Bo-Yuan Ning**, Hezhu Shao, Li Zhao, Xi-Jing Ning, Jun Zhuang*, Vol.**82**, No.**104602**, **Journal of The Physical Society of Japan** (2013).
- * Physical mechanisms for the unique optical properties of chalcogen-hyperdoped silicon, Hezhu Shao, Yuan Li, Jinhu Zhang, **Bo-Yuan Ning**, Wenxian Zhang, Xi-Jing Ning, Li Zhao, Jun Zhuang*, Vol.**99**, No.**46005**, **Europhysics Letters** (2012).

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- * Manipulating dipolar and spin-exchange interactions in spin-1 Bose-Einstein condensates, **Bo-Yuan Ning**, S. Yi, Jun Zhuang, J. Q. You, Wenxian Zhang*, Vol.**85**, No.**053646**, **Physical Review A** (2012).
- * Enhancement of spin coherence in a spin-1 Bose-Einstein Condensate by dynamical decoupling approaches, **Bo-Yuan Ning**, Jun Zhuang, J. Q. You, Wenxian Zhang*, Vol.**84**, No.**013606**, **Physical Review A** (2011).
- * Realization of quantum single pendulum on macroscopic level, **Bo-Yuan Ning**, Jun-Shan Ma, Jun Zhuang, Xi-Jing Ning*, Vol.**59**, No.**1456**, **Acta Physica Sinica** (2010). (undergraduate period)

Conference Talk

- * DIA: A new method to predict phase transition and EOS of solid materials, Annual Meeting of the CPS, SUSTech, Shenzhen, China, Nov. 18-20, 2022.
- * Direct integral approach to many-body partition function, Mainz Materials Simulations Days 2019, Max Planck Institute for Polymer Research, Mainz, Germany, Jun. 5-7, 2019.
- * Manipulating dipolar and spin-exchange interactions in spin-1 Bose-Einstein condensates, 43rd Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Orange County, California, USA, Jun. 4-8, 2012.
- * Spin decoherence in a spinor Bose-Einstein condensate modulated by optical Feshbach resonant pulses, Annual Meeting of the CPS, Zhejiang University, Hangzhou, China, Sep. 15-18, 2011.

Computer Tools

Numerical Calculation *Fortran, CUDA Fortran (GPU-based)*

Software Packages *LAMMPS (Classic MD)
Quantum Espresso, VASP (First-Principles)*

OS *Linux
(Shell-Script Language)*

Data Process *Python Scientific Ecosystem:
Numpy, Scipy, Matplotlib, Sympy, Pandas, openpyxl, Mayavi*

Database: *MongoDB (with pymongo)*
GUI APP: *Qt5 (with PyQt5)*

Others *Emacs Ecosystem
(Elisp Language)*

English Capability

TOFEL 95 (/120)
CET6 569 (/710)
CET4 637 (/710)

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